MIL-S-30506A
15 September 1964
SUPERSEDING
MIL-S-30506(NAVY)
18 April 1963

MILITARY SPECIFICATION

SHACKLES, SHACKLE FITTINGS, AND SWIVELS, BUOY

This specification has been approved by the Department of Defense and is mandatory for use by the Department of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 Scope.- This specification covers shackles, swivels, and shackle pins and split keys, used as chain fittings on Aids to Navigation buoys. Also included are specifications on the "Modeer" shackle, used in some locations for handling buoy chain in lieu of chain slings.

1.2 Classification.- The following classifications do not apply to the modeer shackle.

1.2.1 Shackles and swivels.- Shackles and swivels shall be of the following classes and sizes, as stipulated (see 6.1) (size is defined as bar size or dimension "A" of the shackles and swivels; see Figures 1, 6, and 7):

	Shackles (inches)	Swivels (inches)
1st Class	2	2
2nd Class	1-3/4	1-3/4
3rd Class	1-1/2	1-1/2
4th Class	1	1-1/4
Special	2-1/2	———

In addition, swivels shall be of the following styles, as stipulated (see 6.1):

Upset Head
Threaded Nut

1.2.2 Pins.- Pins for shackles shall be of the following types as stipulated (see 6.1).

Type I Split Key
Type II Rivet Pin

1.2.3 Split Keys.- Split keys shall be of the following classes, as stipulated (see 6.1):

1st, 2nd, and 3rd Class
4th Class
Special

FSC 4030

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2. APPLICABLE DOCUMENTS

2.1 The following military specifications, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner

Military

MIL-S-15083	Steel Castings
MIL-S-16974	Steel Bars, Billets, Blooms and Slabs; Carbon and Alloy (For Reforging or other Operations before Heat Treatment)

STANDARD

Military

MIL-STD-129	Marking for Shipment and Storage

(Copies of specifications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.-

3.1.1 Shackles, swivels, and pins.- Shackles, swivels, and pins shall be made of either forging quality steel conforming to Specification MIL-S 16974 or of steel casting in conformance with class 70-36 of Specification MIL-S-15083.

3.1.2 Split keys.- Split keys shall be made from Hot Rolled Flat Bars of 1010 steel (annealed if necessary) conforming to Specification MIL-S-16974.

3.1.3 The contractor shall furnish an affidavit either from the manufacturer or a reputable engineering laboratory to the effect that the material has been tested and found to be in accordance with the requirements of MIL-S-15083 or MIL-S-16974, as applicable.

3.2 Design and dimensions.- Shackles, swivels, pins, and split keys shall conform in design and dimensions to Figures 1 through 9 and Tables 1 through 6. Figure 6 shows the swivel with an upset head whereas Figure 7 shows the swivel with shank fitted with a threaded nut and then electric welded to secure same. Figure 5 shows shackles when assembled with split key pins or rivet pins.

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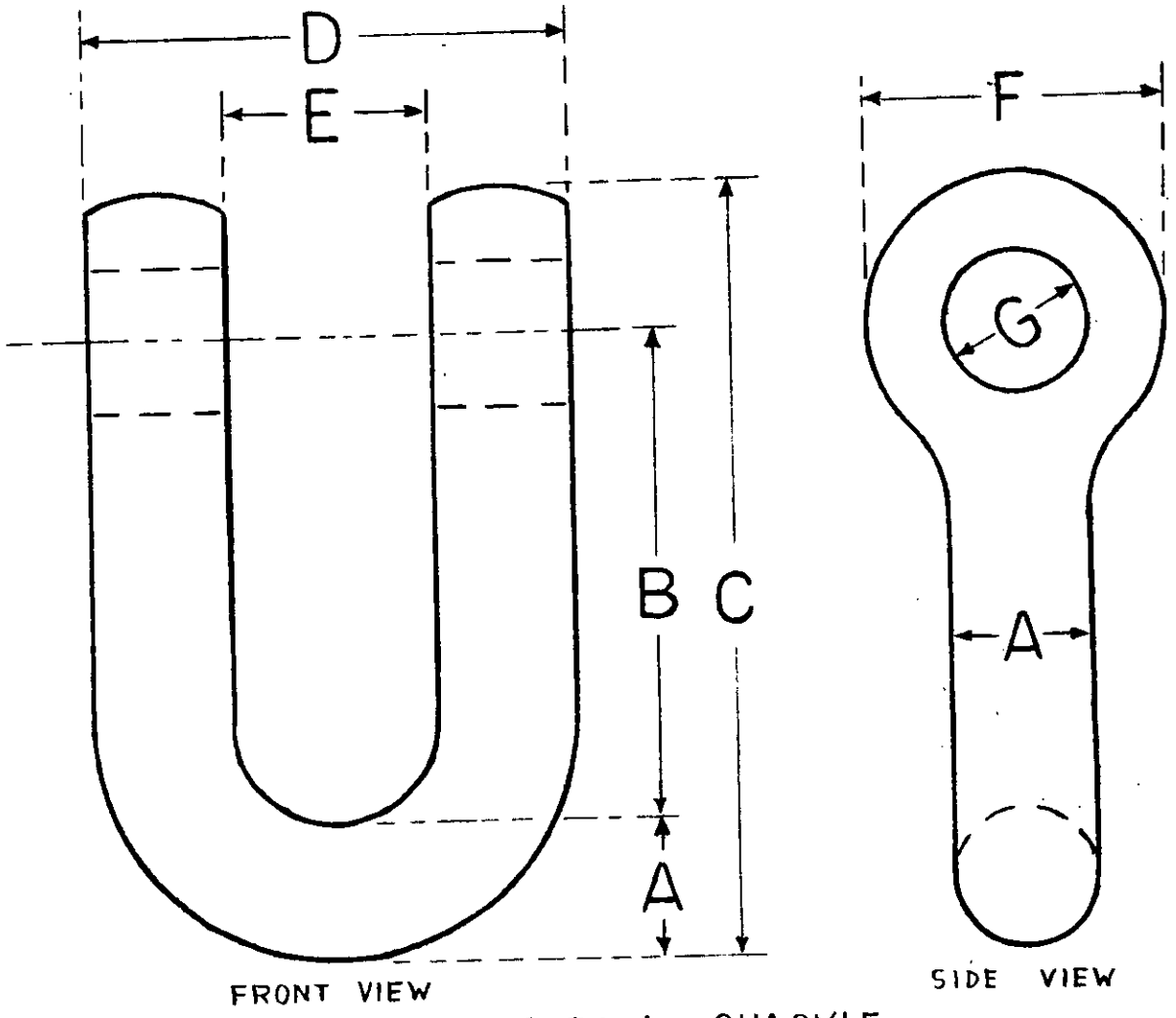


FIGURE 1 SHACKLE

	A	B	C	D	E	F	G
1st Class	2	7-3/4	11-7/8	7	3	4-1/4	2-1/16
2nd Class	1-3/4	6-7/8	10-1/2	6-1/4	2-3/4	3-3/4	1-13/16
3rd Class	1-1/2	5-1/4	8-3/8	5-1/4	2-1/4	3-1/4	1-9/16
4th Class	1	4-1/8	6-1/8	4-1/4	2-1/4	2	1-1/16
Special	2-1/2	9	14-1/4	11	6	5-1/2	2-9/16

TABLE I SHACKLE DIMENSIONS (IN INCHES)

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	1st, 2nd, and 3rd Class	4th Class	Special
R	3/16	1/8	1/4
S	9/16	9/16	9/16
T	15/16	13/16	1-1/16
U	1-1/4	7/8	1-1/4
V	3/8	3/8	3/8
W	3/8	1/4	1/2
X	4	4	4-1/2

TABLE II SPLIT KEY DIMENSIONS (IN INCHES)

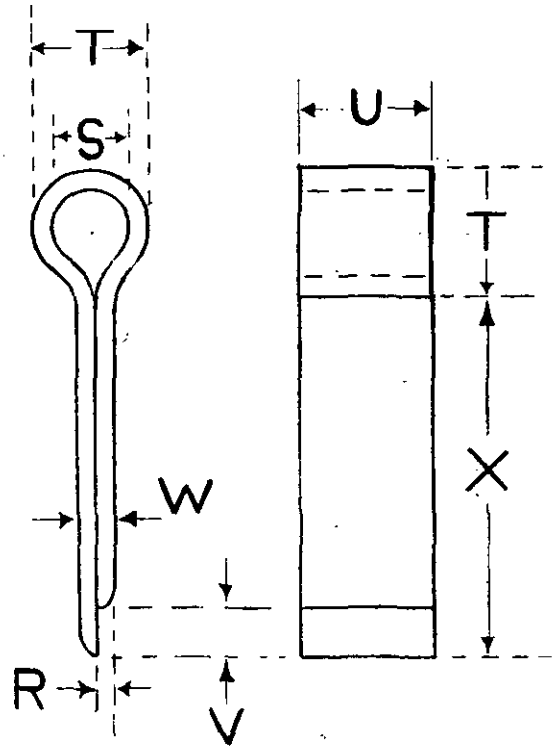


FIGURE 2 SPLIT KEY

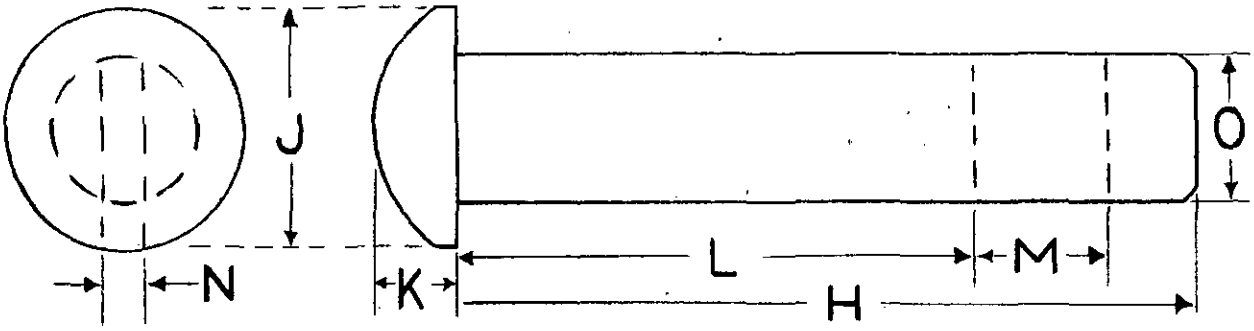


FIGURE 3 SPLIT KEY PIN

	H	J	K	L	M	N	O
1st Class	9-1/2	3	1-1/8	7-1/16	1-7/16	7/16	2
2nd Class	8-3/4	2-3/4	1	6-5/16	1-7/16	7/16	1-3/4
3rd Class	7-3/4	2-1/2	7/8	5-5/16	1-7/16	7/16	1-1/2
4th Class	6	1-5/8	5/8	4-5/16	1-1/16	5/16	1
Special	13-5/8	3-1/2	1-3/8	11-1/8	1-7/16	9/16	2-1/2

TABLE III SPLIT KEY PIN DIMENSIONS (IN INCHES)

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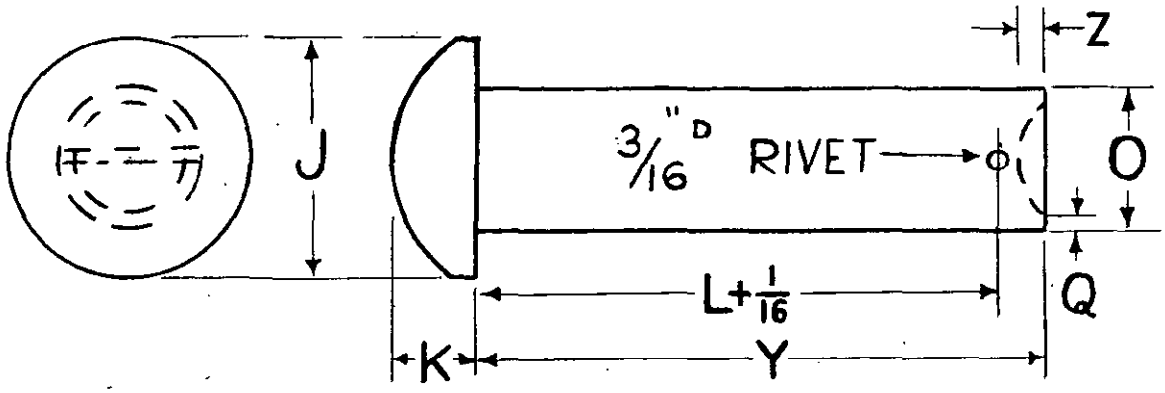
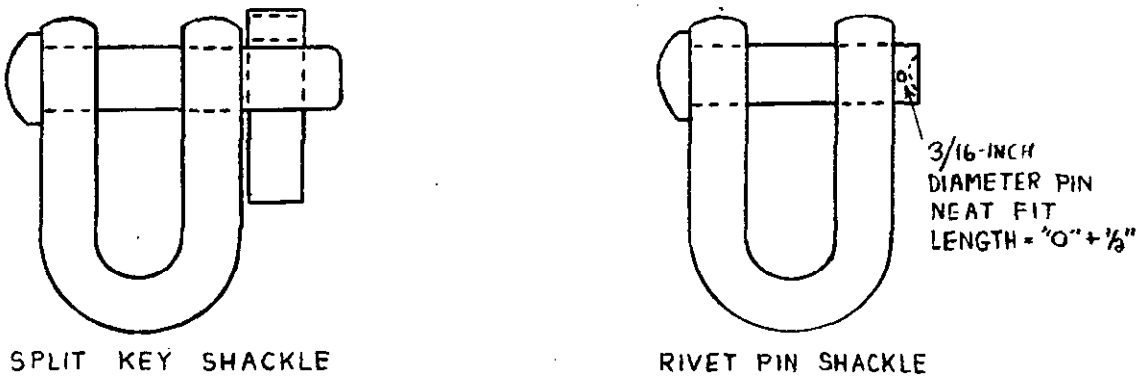


FIGURE 4 RIVET PIN

	J	K	L	O	Q	Y	Z
1st Class	3	1-1/8	7-1/16	2	3/16	8	3/8
2nd Class	2-3/4	1	6-5/16	1-3/4	3/16	7-1/8	3/8
3rd Class	2-1/2	7/8	5-5/16	1-1/2	3/16	5-7/8	1/4
4th Class	1-5/8	5/8	4-5/16	1	1/8	4-7/8	1/4

TABLE IV RIVET PIN DIMENSIONS (IN INCHES)



SPLIT KEY SHACKLE

RIVET PIN SHACKLE

FIGURE 5 SHACKLE ASSEMBLIES

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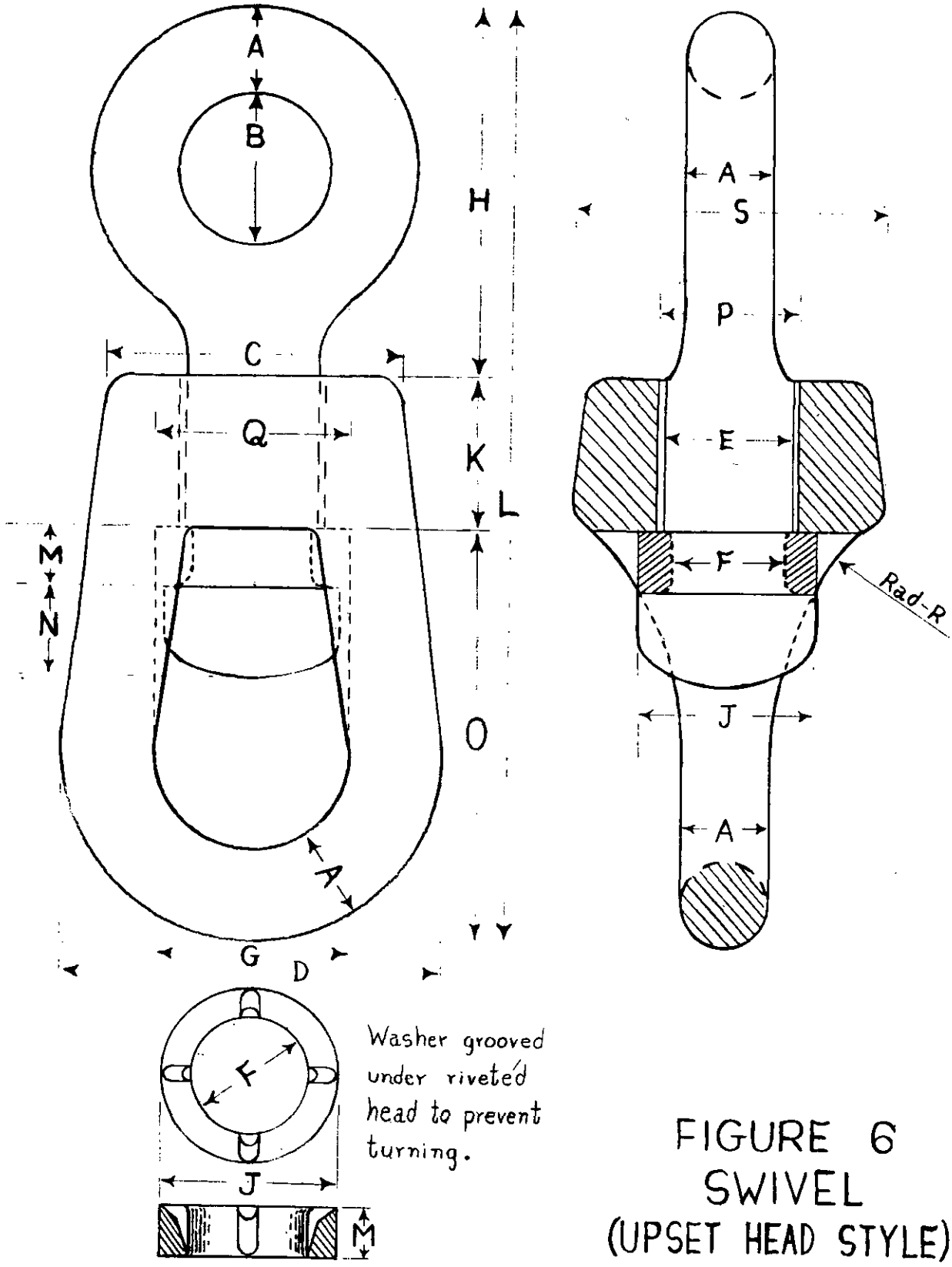


FIGURE 6
SWIVEL
(UPSET HEAD STYLE)

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	1st Class	2nd Class	3rd Class	4th Class
A	2	1-3/4	1-1/2	1-1/4
B	3	2-3/4	2-1/2	2-1/4
C	6	5	4-1/2	4
D	8	7	6	5
E	3	2-1/2	2-1/4	2
F	2-3/4	2-1/4	2	1-3/4
G	4	3-1/2	3	2-1/2
H	7-3/4	7	6-1/4	5-1/4
J	4	3-1/4	3	2-3/4
K	3	2-3/4	2-1/2	2
L	19-3/4	17-3/4	15-3/4	13
M	1	1	7/8	7/8
N	2	1-3/4	1-1/2	1-1/4
O	9	8	7	5-3/4
P	3-1/4	2-3/4	2-1/2	2-1/4
Q	4-3/8	3-5/8	3-1/4	3
R	6	5	4	3-1/2
S	6-3/4	5-3/4	5	4-1/2

TABLE V
SWIVEL DIMENSIONS (IN INCHES)
(UPSET HEAD STYLE)

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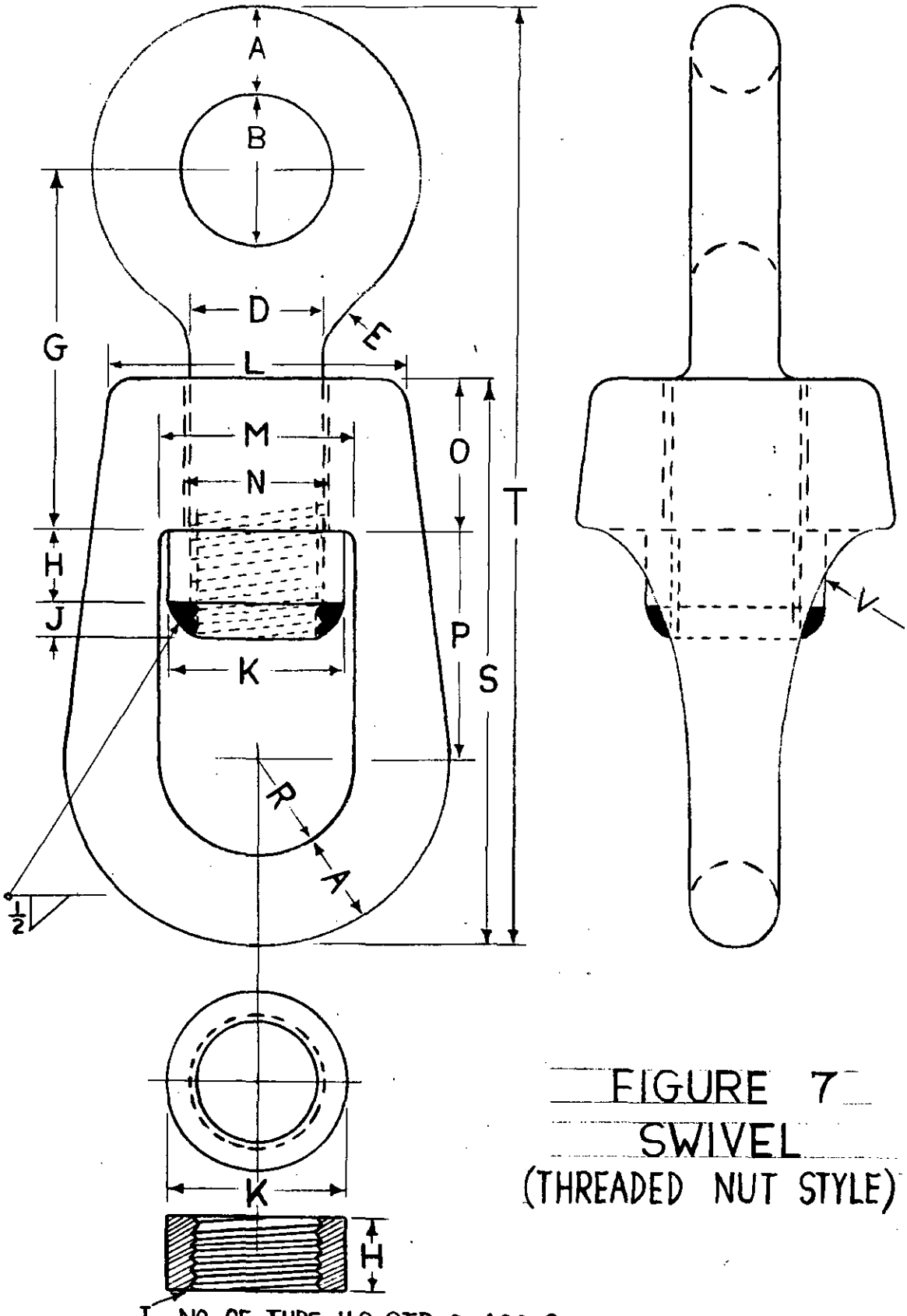


FIGURE 7
SWIVEL
(THREADED NUT STYLE)

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	1st Class	2nd Class	3rd Class	4th Class
A	2	1-3/4	1-1/2	1-1/4
B	3	2-3/4	2-1/2	2-1/4
D	3	2-1/2	2-1/4	2
E	1-5/8	1-3/8	1-1/4	1-1/8
G	7-1/4	6-5/8	6	4-7/8
H	1-1/2	1-1/2	1-1/4	1-1/4
I	3-1/2	4	4-1/2	4-1/2
J	1/2	1/2	1/2	1/2
K	4	3-1/4	3	2-3/4
L	6	5	4-1/2	4
M	4-3/8	3-5/8	3-1/4	3
N	3-1/4	2-3/4	2-1/2	2-1/4
O	3	2-3/4	2-1/2	2
P	4-3/4	4-3/8	3-7/8	3
R	2-1/4	1-7/8	1-5/8	1-1/2
S	12	10-3/4	9-1/2	7-3/4
T	19-3/4	17-3/4	15-3/4	13
V	9	8	7	6-1/2

TABLE VI
SWIVEL DIMENSIONS (IN INCHES)
(THREADED NUT STYLE)

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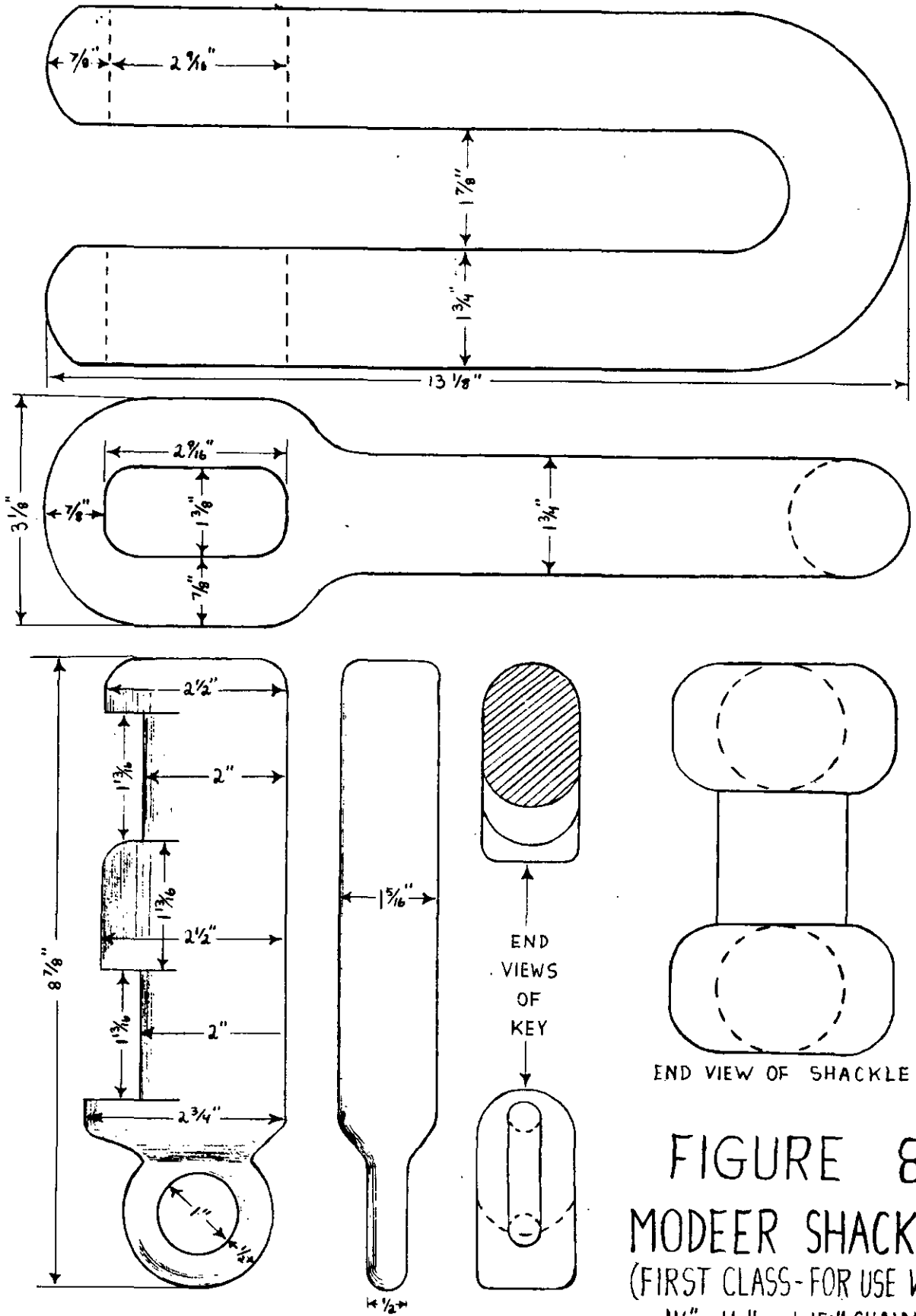


FIGURE 8
MODEER SHACKLE
(FIRST CLASS - FOR USE WITH 1¼" - 1½" and 1⅝" CHAIN)

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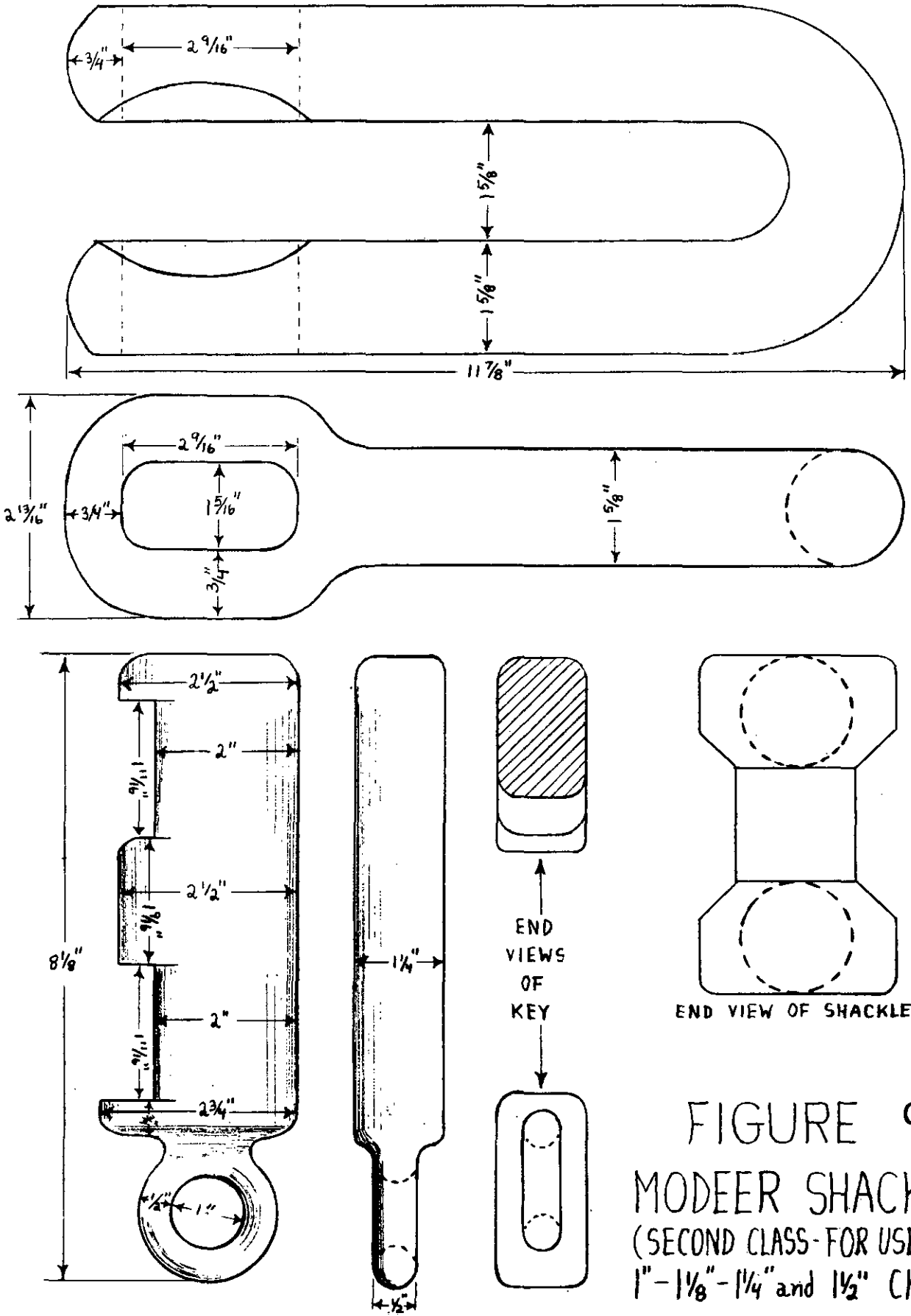


FIGURE 9
MODEER SHACKLE
(SECOND CLASS-FOR USE WITH 1"-1⅛"-1¼" and 1½" CHAIN)

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3.2.1 Dimension tolerances.- Dimension tolerances shall be as follows:

Fig. 1 Shackle body:	variation of .063-inch.
Shackle holes to take split keys and round pins:	variation of .03-inch oversize
Fig. 2 Bars used to make split keys:	variation of .01-inch on thickness and .06-inch oversize on width.
Figs. 3 & 4 Round Pin Body:	variation of .03-inch oversize.
Figs. 6 & 7 Swivel Body:	variation of .063-inch.

3.3 Weight.- The weights of shackles and swivels shall be as shown in Table VII.

3.4 Proof Loads.- Shackles and swivels shall be capable of withstanding the proof loads specified in Table VII without developing surface rupture or defects which would interfere with serviceability or prevent disassembly where applicable. The static testing machine used shall be approved by the procuring agency.

3.5 Breaking Loads.- Shackles and swivels, with their applicable pins, keys, or bolts, shall be able to withstand, without breaking, the breaking loads shown in Table VII.

	Proof Load	Breaking Load	Weight Each (Approx)
Shackles			
1st Class	113,100	226,200	40
2nd Class	86,400	172,800	27
3rd Class	63,350	126,700	16
4th Class	28,500	57,000	6
Special	176,400	352,800	75
Swivels			
1st Class	81,500	163,000	78
2nd Class	59,000	118,000	52
3rd Class	45,000	90,000	39
4th Class	30,000	60,000	19

Table VII Physical Test Values

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3.6 Workmanship.- All shackles, swivels, pins, and keys shall be smooth and free from flaws, seams, cracks, and other injurious defects which may affect their serviceability and handling. Screw threads shall be true to form, clean cut, and free from injurious defects.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot.- For the purpose of inspection and testing a lot shall consist of all items of the same type, class, and style offered for acceptance at the same time.

4.3 Sampling.-

4.3.1 Shackles, swivels, and pins.- Sample shackles and swivels for inspection and for breaking and proof tests shall be selected from each lot in accordance with Table VIII. Samples used for proof test may be re-used for the breaking tests. Pins, when procured alone, shall be subjected to this same sampling plan.

Number of items in lot	Inspection	Proof	Breaking
1 to 10	All	2	1
11 to 30	All	2	1
31 to 50	30	2	1
51 to 100	42	2	2
101 to 200	55	6	3
201 to 400	60	8	4
401 to 800	65	10	5
801 to 4,000	70	10	5
4,001 to 10,000	75	10	5

Table VIII Sampling, Swivels, and Pins

4.3.2 Split keys.- Samples for inspection of split keys shall be selected from each lot in accordance with Table IX.

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Number of keys in lot	No. of samples	Acceptance No.
Up to 500	20	1
500 - 800	25	2
800 - 1,300	35	2
1,300 - 3,200	50	4
3,200 - 8,000	75	6
8,000 - 25,000	100	8
25,000 - 100,000	150	11

Table IX Split Key Sampling

4.4 Inspection.- Samples selected in accordance with 4.3.1 and 4.3.2 shall be inspected by the Government inspector to determine conformance with paragraphs 3.2 and 3.6 of this specification.

4.5 Proof tests.- Sample shackles, swivels, or pins set in the applicable shackles, selected in accordance with 4.3, shall be subjected to the applicable proof loads shown in Table VII, either singly or in series. The diameter of the attachments used in proofing shackles shall be not greater than the pin diameter of the shackle being tested, and the application of the proof load shall be at center of the pin. After releasing the proof load, each shackle or swivel shall be examined and disassembled to determine conformance with 3.3.

4.6 Breaking tests.- After proof testing, sample shackles, swivels, or pins set in the applicable shackles, selected in accordance with 4.3, shall be subjected to the breaking loads shown in Table VII, to determine conformance with 3.4. The diameter of the attachment used in the breaking tests of shackles shall be not greater than the pin diameter of the shackle being tested; the load application shall be at the center of the pin.

4.7 Rejections.-

4.7.1 Inspection.- If one or more of the sample shackles, swivels, or pins inspected in accordance with 4.4 fail to conform with 3.2 and 3.5, the entire lot shall be rejected. In inspection of split keys, if the cumulative number of defective samples is equal to or less than the acceptance number shown in Table IX, the lot may be accepted. If the number of defective samples exceeds the acceptance number, a second set of samples shall be selected, in an amount doubling that taken originally, and the acceptance number shall also be doubled. If the number of defective samples exceeds the second acceptance number the lot shall be rejected.

4.7.2 Proof test.- If any sample proofed in accordance with 4.5 fails to pass the proof test, all items in the lot shall be proofed and those failing to pass the proof test shall be rejected.

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4.7.3 Breaking test.- If any sample fails to withstand the breaking load when tested in accordance with 4.6, double the number of the original samples shall be subjected to the breaking test. If any retest sample fails the breaking test, the lot shall be rejected.

4.8 Disposition of test samples.- Shackles, swivels, or pins subjected to the breaking tests of 4.6 shall be destroyed. Samples passing the proof tests of 4.5 shall be retained.

4.9 Inspection of preparation for delivery.- The packing and marking of the equipment shall be inspected to determine conformance with the applicable requirements of section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Packing shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A.- The shackles, swivels, pins and keys shall be packed in a cleated-plywood wood box or a nailed and lock-corner wood box conforming to Specifications PPP-B-601 (overseas type) and PPP-B-621 (class 2), respectively. The contents shall be cushioned, blocked, and braced to prevent movement and the boxes shall be closed and strapped in accordance with appendix to the applicable box specification; flat steel strapping shall be class B.

5.1.2 Level B.- The shackles, swivels, pins, and keys shall be packed as specified for level A, except that the boxes shall be domestic type or class, as applicable, and the class B strapping requirement does not apply.

5.1.3 Level C.- The shackles, swivels, pins, and keys shall be packed in a manner that will insure arrival at destination in satisfactory condition and be acceptable to the carrier at lowest rates. Shackles shall be bundled in the amounts shown.

1st Class - 4 per bundle
2nd Class - 5 per bundle
3rd Class - 5 per bundle
4th Class -10 per bundle
Special - 3 per bundle

5.2 Marking.- In addition to any special marking required by the contract or order, unit packages and shipping containers shall be marked in accordance with the requirements of Standard MIL-STD-129.

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6. NOTES

6.1 Intended use.- Fittings covered by this specification are intended for use on navigational buoys, moorings for pontoons, causeways, and barges. The fittings shall not be used for applications that involve ship moorings.

6.2 Ordering data.- Procurement documents should stipulate the following:

(a) Title, number, and date of this specification.
(b) Type, class, or style required (see 1.2).
(c) Quantity required.
(d) Federal stock numbers.
(e) Level of packing (see 5.1).

6.3 The 3/16-inch diameter pin shown in Figure 5 is only required when the complete rivet pin shackle assembly is ordered, and is NOT required when the round shackle pin only is ordered. The 3/16-inch pin shall be bent on both ends to prevent loss when assembled.

NOTICE. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodians:
Navy - Bureau of Yards and Docks
Air Force - Mobile Air Materiel Area
Army - Mobility Command

Preparing activity:
U. S. Coast Guard

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